

'Auslan Alexa': A case study of VR Wizard of Oz prototyping for requirements elicitation with Deaf participants

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ABSTRACT

We used VR as a speculative design tool to explore how Deaf people might interact with a sign language personal assistant through a Wizard of Oz pilot study. This allowed us to elicit concrete design requirements from potential users of the proposed personal assistant, and shows the potential of VR as a requirements elicitation tool.

Index Terms: Human-centered computing—Human-computer interaction (HCI)—Interaction paradigms—Virtual reality; Human-centered computing—Accessibility

1 INTRODUCTION

Personal assistants (PAs) are commercially available for mainstream spoken languages, but inaccessible to users of sign languages. To address this access gap, sign language PAs (SLPAs) must be developed, which requires significant resources and technical AI development. Without sufficient user input into the design, they may not meet user needs. In this project, we designed a VR 'smart' apartment with a Wizard of Oz activated SLPA. This paper presents a pilot study we undertook to evaluate the feasibility of VR as a requirements elicitation tool when working with Deaf people.

2 BACKGROUND

2.1 Deafness

Upper case Deaf indicates cultural/linguistic membership of a Deaf community. In Australia, Deaf people are a linguistic minority embedded in mainstream English-using culture [7], so many Deaf people have some level of bilingualism. In contrast, Hard of Hearing people (HoH) generally belong to hearing culture. Deaf and HoH people may choose to use cochlear implants or hearing aids.

2.2 Personal Assistants

62.8% of Deaf and HoH people in [5] were interested in a PA they could sign to; and 70.9% were interested in a PA which signed back. However, 80.2% of those surveyed had never used a PA; they had to imagine interactions based on video of hearing people interacting with a speaking PA. This demonstrates a need to provide Deaf people experience of interacting with a PA as part of the design process.

2.3 Wizard of Oz (WoZ)

In WoZ prototyping, a human operator secretly executes back end functionalities as a means to observe interaction with a prototype [3].

2.4 VR and Requirements Elicitation

As a requirements elicitation tool, [1] found VR resulted in more and qualitatively better requirements. Immersion, presence and intuitive locomotion positively impacted participants' experience.

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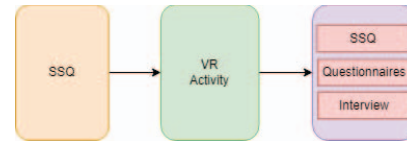


Figure 1: Experimental Procedure

Table 1: Questionnaires

Phase	Questionnaire Sections
1	VR Locomotion Experience Questionnaire (VRLEQ) [2] Presence and Immersion questionnaire [6]
2	Perceived Usability of 'Auslan Alexa' Command Usage Frequency [5] Command Execution Method Preferences [5] Wakeup Method and Location Preferences [5] Privacy and Security [5]

3 METHODOLOGY

A virtual environment was created based on 'HQ Archviz Modern Studio Apartment' by Next Level 3D Studio. A PA screen was designed in Solidworks, taking into account research on reading [4] and health and safety [9]. We used a HTC Vive Pro head-mounted display. Marching-in-place locomotion was implemented through Natural Locomotion software using HTC Vive Trackers worn around the ankles for feet tracking and HTC Vive Controllers for mapping forward movement. This was selected as it minimised cybersickness, promoted immersion and presence, was easy to understand, and left participants' hands free for signing.

The study had two phases: Phase 1 to refine the VR experience (n=3 Hearing, 1 HoH), and Phase 2 to obtain requirements for a SLPA (n=3 Deaf). The procedure and questionnaires for each phase are given in Fig. 1 and Table 1. During the VR activity, participants gave commands (verbally in Phase 1; signed and verbalised by an Auslan interpreter in Phase 2) which the WoZ operator executed. The open-ended interviews focused on the VR experience in Phase 1 (English); and the PA in Phase 2 (via Auslan interpreter). HoH and Deaf participants were also asked about disruptions to hearing aids from the headset. WoZ was revealed late in the interview.

4 RESULTS

4.1 Cybersickness

Only one Phase 1 participant showed significant motion sickness on the SSQ. Interviews revealed flaws in locomotion (no speed regulation, floating sensation) that might cause cybersickness, which were addressed between phases. Phase 2 participants showed more variation in SSQ scores, including one participant with severe cybersickness after a prolonged VR experience due to faulty locomotion.

4.2 Locomotion

Locomotion was given a SUS score of 85.75 in Phase 1 VRLEQ, showing high satisfaction, although competence and challenge were in the moderate range. In interviews, participants mentioned movement not translating to the VR space, and "getting stuck" in furniture.

4.3 Presence and Immersion

Phase 1 questionnaires and interviews showed that presence and immersion succeeded, with graphics and imagery praised. Participants reported immersion breakages: 1) Malfunctions of locomotion; 2) Hearing the WoZ keyboard; and 3) The wire touching the participant.

4.4 Impact of Headset on Hearing Aids

The Phase 1 HoH participant reported her hearing aid's phone mode getting triggered once (assumed to be due to the headset's magnets).

Two Phase 2 participants wore hearing aids. One reported cybersickness symptoms when asked about impacts to his hearing aids; the second mentioned hearing a very slight interference.

4.5 Personal Assistant Device and Wizard of Oz

Phase 1 participants provided positive feedback about their interactions with the PA device and the WoZ procedure. Their feedback was used to improve the VR prototype's usability (increased PA text size, relocated PA screen for increased visibility, smoother locomotion), expand the PA's abilities (e.g. video call, music, timers), and improve the WoZ procedure for Phase 2 (quieter keyboard).

4.5.1 Personal Assistant Usability

Phase 2 SUS results showed general dissatisfaction with the PA. Interviews revealed that participants wanted more functionality, and the WoZ operator's "close enough" substitutions had been insufficient. The general nature of responses in the prototype (e.g. "Item added to shopping list"), which had been chosen to support WoZ, was unsatisfying. When participants gave multiple similar commands for which only one response had been programmed (e.g. weather reports in multiple cities), participants commented on lack of accuracy in responses.

4.5.2 Personal Assistant Requirements

Deaf participants provided feedback and requirements about: PA appearance, size and clarity; desired functionality; connection to smart watches for notifications; PA wake-up methods; PA response languages; PA placement considerations; privacy and security concerns; customisability; and potential usage scenarios.

4.5.3 Observed Interactions

Deaf participants were very polite in PA interactions (e.g. signing "please" with each command). They also tried different ways of expressing commands that were not understood (e.g., fingerspelling when a sign was "not recognised").

4.5.4 Wizard of Oz

When Phase 2 participants were told about the WoZ, two participants said they had guessed. However, this did not seem to negatively influence the use of the VR prototype as a requirements elicitation tool, as noted by the depth and breadth of feedback provided and they way they interacted with the device.

5 DISCUSSION AND CONCLUSION

The use of a VR prototype enabled Deaf participants to experience and provide feedback on a device which does not yet exist, and which cannot exist without substantial time and monetary investment: a SLPA. The immersive properties of VR supported realistic interactions with the prototype. Qualitative interviews and observations of Deaf people's interactions with the device allowed us to unpack quantitative usability scores to understand what motivated

both positive and negative results. We were also able to catalogue as-yet-unknown requirements about the device's functionality and Deaf people's preferences for interaction. Limitations of the PA prototype (e.g. generic response states) may have negatively impacted participants' experience of the virtual PA, but became sources of requirements. Difficulties encountered with the VR system, especially locomotion and cybersickness, negatively impacted participants' opinions of the prototype PA in ways that did not produce requirements data. This is an important factor to consider in requirements elicitation.

The primary limitation of this research was the small number of participants. This is appropriate in qualitative research such as requirements elicitation. However, this means quantitative scores obtained during this study are indicative only, not generalisable.

Future research exploring VR for requirements elicitation could work with other user groups, particularly other minority and disabled user groups.

Future work in the umbrella project [8] will involve implementing an Auslan PA, incorporating the requirements identified in this study.

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